FACE MASK DETECTION CODE FLOW

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MODEL TRAINING

DATA COLLECTION

01

- Images are collected manually
- Labelled and annotated using annotation tools like Label Studio

02

- Required images are available in web sources like kaggle
- Images are classified into class wise folders, so annotations are not necessary

MODEL TRAINING

DATA PREPROCESSING

- Input and output are converted to an array format i.e. Data and labels are converted to an array
- Convert the labels(classes) to numbers using the one-hot encoding method

MODEL TRAINING

DATA SPLITTING(TRAIN & TEST DATA)

- Split the test and training data using test_train_split and store it in the appropriate input and output variables of test and train

MODEL TRAINING

TRANSFER LEARNING

- We transfer the pre-trained model weights to the basemodel and mention whether to include top layer or not
- Input shape is set based on the shape of the input given to the model
- Base model output is given as input to the headmodel

MODEL TRAINING

MODEL CREATION

- After the above step the below processes are done to the head model:
    1. AveragePooling
    2. Flatten
    3. Dense layer(fully connected layer)
    4. Drop out(reducing the model accuracy by 0.5)
    5. Dense layer
- Defining the model input as 'basemodel.input' and output as the 'head model'

MODEL TRAINING

MODEL CREATION

- Loop over all the basemodel layers using for loop and freeze them so they are not updated during training process
- Define the optimizer with the required parameters like learning rate

MODEL TRAINING

MODEL CREATION

- Define the model compile parameters like metrics and loss
- once the model compile we will get the loss and accuracy evaluation
- Train the model using fit method

MODEL TRAINING

PREDICTION

- Predict the output of the trained model using predict method by passing the test data
- Argmax is used to find the max index of the predicted array values

MODEL TRAINING

CLASSIFICATION REPORT

- Classification report of the model is generated by passing the actual and predicted values

MODEL TRAINING

SAVING THE MODEL

- Best model saved using save method with a required file format

APPLICATION FLOW

FACE DETECTION AND PREDICTION

- Frame of the image (in which people face is captured) is given as input to the facenet model
- Facenet model will crop the people face from the input
- Using for loop is used to detect all the faces in the particular frame
- If there is a single face then only prediction for the model will be done
- Available faces in the frame will be stored in the face list

APPLICATION FLOW

LOADING PRE-TRAINED MODEL

- Load the face detector (facenet) model by defining the prototxt path (faces image path) and weights path
- Load the face mask detector(masknet) model by passing the trained model

APPLICATION FLOW

INITIALIZE VIDEO SCREEN

- Start the videostream using VideoStream method
- Sleep timer is set to 2.0

APPLICATION FLOW

EXTRACT FRAME FROM VIDEO

- Loop over the frames from video using While loop
- Get the frame from video and resize it using imutils and store it to frame list
- Call the detection and prediction function to detect whether the person is wearing mask or not
- Save the locations and predictions from the function to locs and preds list

APPLICATION FLOW

PREDICTION RESULTS

- For loop is used to draw box in the detected face locations with the help of the values stored in locs and preds list
- Prediction results like mask or without mask will be saved to pred

APPLICATION FLOW

CLASS IDENTIFICATION

- Labelling the face as with or without mask with the mentioned color

APPLICATION FLOW

DISPLAYING LABEL WITH BOX ON FRAME

- if-elif loop and cv2 is used to display the label and box on frame
- if loop is used to verify the label value and cv2 is used to display the text for the particular label and rectangle box is drawn based on the locations of the detected face

APPLICATION FLOW

CLOSE CAMERA

- if the user press a particular key('q') then the camera will be closed (i.e. recording will be stopped)

THANK YOU